Summer Research Fellowship Programme, 2026

A Study of Galton–Watson Branching Processes and Their Long-Term Behaviour

Eight-Week Progress Report

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Acknowledgements

I would like to express my sincere gratitude to my guide, **Dr. Moumanti Podder**, Department of Mathematics, IISER Pune, for introducing me to the theory of branching processes and for her valuable guidance throughout this project. Her suggestions on the choice of topics and reading material greatly influenced the direction of my study and helped me develop a deeper appreciation for probabilistic methods.
I am grateful to the **Indian Academy of Sciences (IASc)**, the **Indian National Science Academy (INSA)**, and **The National Academy of Sciences, India (NASI)** for providing me with the opportunity to participate in the Summer Research Fellowship Programme. The programme offered an excellent environment to explore an area of mathematics beyond the regular curriculum and to engage with contemporary research ideas.
I would also like to thank the faculty members of the **Indian Statistical Institute, Kolkata**, whose courses in probability theory and related subjects provided the background necessary for undertaking this project.
Finally, I am thankful to my family and friends for their constant encouragement and support throughout the course of this work.

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Chapter 1

Introduction

Branching processes are among the simplest stochastic models used to describe the evolution of populations whose individuals reproduce independently. Originally introduced by Francis Galton and Henry Watson in the nineteenth century to study the extinction of family names, they have since found applications in diverse areas such as population biology, epidemiology, genetics, nuclear chain reactions, computer science, and the spread of information through networks.
Despite their simple formulation, branching processes exhibit a rich variety of long-term behaviours. Depending on the underlying offspring distribution, a population may become extinct, remain stable for long periods, or experience sustained growth. Understanding how these outcomes arise and how they depend on the parameters of the model forms one of the central themes of branching-process theory.
The objective of this report is to study the Galton–Watson branching process and trace the progression from its basic probabilistic structure to its asymptotic behaviour. The first part develops the fundamental framework of the model, introducing the branching property, moment calculations, and generating functions as the principal analytical tool. These ideas culminate in the study of extinction probabilities, where generating functions reveal a deep connection between long-term behaviour and fixed-point equations.
The second part focuses on questions that arise beyond extinction. While extinction probabilities determine whether a population survives, they provide limited information about the size and structure of surviving populations. This motivates the introduction of normalized processes, martingale methods, conditioned limit laws, and asymptotic techniques. The report then studies stronger asymptotic results, including martingale convergence, Seneta normalization, geometric convergence of generating functions, conditioned limit laws, the Q-process, and diffusion approximations.
Throughout the report, emphasis is placed on the ideas and motivations underlying the theory rather than on technical proofs. Particular attention is given to recurring themes such as recursion, generating functions, conditioning, scaling, and limit laws, many of which reappear throughout modern probability theory and related areas of mathematics.

Chapter 2

Foundations: Modelling Population Growth

The study of branching processes originated from attempts to understand the evolution of populations whose individuals reproduce independently from one generation to the next. Questions concerning the survival or extinction of family names, the spread of epidemics, the growth of biological populations, and the propagation of information through networks can all be viewed through a common mathematical framework. The simplest and most influential model in this direction is the *Galton–Watson process*, introduced by Francis Galton and Henry Watson in the nineteenth century while investigating the extinction of aristocratic family names.

2.1 The Galton–Watson Process

Consider a population evolving in discrete generations. Each individual produces a random number of offspring according to a fixed probability distribution

$$P(\text{offspring number} = k) = p_k, \qquad k = 0, 1, 2, \dots,$$

where the offspring numbers of different individuals are assumed to be independent and identically distributed. If Z_n denotes the population size in the n-th generation, then the size of the next generation is obtained by summing the offspring produced by all individuals currently alive:

$$Z_{n+1} = \sum_{r=1}^{Z_n} X_{n,r},$$

where the random variables $X_{n,r}$ are independent and have common distribution p_k.
This recursive relation captures the fundamental idea of random growth. The future evolution of the population depends only on its present size and not on the manner in which that size was attained. Consequently, the sequence $Z_{nn \geq 0}$ forms a Markov chain on the non-negative integers. The state 0 plays a special role: once the population reaches zero, no further offspring can be produced and extinction becomes permanent.
An important structural feature of the model is the independence of family lines. A process initiated with several ancestors can be viewed as the sum of independent copies of a process started from a single ancestor. This branching property lies at the heart of many theoretical developments and distinguishes branching processes from more general stochastic models.

2.2 Moments and Population Characteristics

While the complete distribution of Z_n contains detailed information about the population, many qualitative features can already be understood through its moments. In particular, the mean population size provides a first indication of whether the process tends to grow, remain stable, or decline over time.
Let m denote the mean number of offspring produced by a single individual. Since every individual reproduces independently according to the same offspring distribution, the expected size of the n-th generation is given by $E[Z_n] = m^n$.
This simple formula reveals a fundamental trichotomy. When $m < 1$, the expected population size decreases geometrically and the process is called **subcritical**. When $m = 1$, the expected population size remains constant and the process is said to be **critical**. Finally, when $m > 1$, the expected population size grows exponentially and the process is termed **supercritical**. Much of the long-term theory of branching processes is organized around these three regimes.
Although expectations provide a useful first approximation, they do not capture the entire behaviour of the process. Two populations may have the same expected size while exhibiting very different levels of variability. For this reason, higher moments such as the variance are also important. The variance of Z_n depends not only on the mean offspring number but also on the variability of the offspring distribution itself. As generations progress, these fluctuations accumulate and play a crucial role in determining the eventual fate of the population.
The moment calculations therefore serve two purposes. First, they **provide quantitative information about the average growth of the population**. Second, they **suggest that expectations alone are insufficient for understanding long-term behaviour**. This observation motivates the search for more powerful analytical tools capable of describing the entire distribution of Z_n, leading naturally to the theory of generating functions.

2.3 Generating Functions as an Analytical Tool

Although the offspring distribution completely specifies a branching process, working directly with infinitely many probabilities quickly becomes cumbersome. A more efficient approach is to encode the entire distribution into a single analytic object known as the *generating function*. This transformation converts probabilistic questions into problems involving functional equations and iteration, making many otherwise difficult calculations tractable.
For an offspring distribution $\{p_k\}_{k\geq 0}$, the associated generating function is defined by

$$f(s) = \sum_{k=0}^{\infty} p_k s^k, \qquad |s| \leq 1.$$

The generating function contains complete information about the offspring law. In particular, moments of the distribution may be recovered through differentiation, while convolutions of distributions become products of generating functions. Since the size of each generation is obtained by combining the offspring contributions of many individuals, generating functions provide a natural language for studying the evolution of the process.
One of the most remarkable features of branching processes is that successive generations correspond to successive compositions of the generating function. If $f_n(s)$ denotes the generating function of the population size in generation n, then

$$f_{n+1}(s) = f(f_n(s)).$$

Consequently, the behaviour of the population after n generations is encoded by the n-fold iterate of f. Questions concerning extinction, growth, and asymptotic behaviour can therefore be reformulated as questions about the iterates of a single function.
This observation marks a major shift in perspective. Instead of studying the random variables $\{Z_n\}$ directly, one studies the sequence of functions $\{f_n\}$. Much of the theory developed in the subsequent sections rests on understanding the fixed points and limiting behaviour of these iterates.

Chapter 3

From Extinction to Limit Laws

3.1 Why Extinction Probabilities Are Not Enough

The extinction probability provides only a partial description of a branching process. By the end of the previous chapter, it was possible to determine whether a population eventually dies out and how this behavior depends on the mean offspring number m. However, extinction is only one aspect of the process. Even when extinction does not occur, many important questions remain unanswered. How large is the surviving population? Does it grow at a predictable rate? Is there a meaningful limiting object that describes its long-term behavior?
The first clue comes from the observation that

$$E[Z_n] = m^n.$$

In the supercritical case $m > 1$, the expected population size grows exponentially. This suggests that the dominant deterministic trend is already known. The natural idea is therefore to remove this growth and investigate whether a nontrivial random limit remains. Such questions lead naturally to the study of normalized versions of the population process and mark the beginning of the asymptotic theory of branching processes.

3.2 Scaling and Long-Term Behaviour

The branching property implies that, conditionally on the current population size, the future population evolves as a collection of independent copies of the original process. As a consequence,

$$E(Z_{n+k} \mid \mathcal{F}_n) = m^k Z_n,$$

where $\mathcal{F}_n$ denotes the information available up to generation n. Dividing by the expected growth factor m^{n+k} suggests introducing the normalized process

$$W_n = \frac{Z_n}{m^n}.$$

The significance of this normalization is that it removes the deterministic exponential growth and isolates the genuinely random component of the population size.

Theorem 3.1. *Suppose $0 < m < \infty$. Then the sequence*

$$W_n = \frac{Z_n}{m^n}$$

forms a nonnegative martingale. Consequently, there exists a random variable W such that

$$W_n \longrightarrow W \qquad \textit{almost surely.}$$

This result provides the first indication that the long-term behavior of the process may be captured by a limiting random variable. The quantity W may be viewed as the eventual random amplitude of population growth after the deterministic factor m^n has been removed. Nevertheless, the theorem leaves open an important question: whether the limit is nontrivial.
The answer is provided under a finite variance assumption.

Theorem 3.2. *Assume that*

$$m > 1, \qquad \mathrm{Var}(Z_1) < \infty,$$

and $Z_0 = 1$. Then W_n converges to W in L^2, and

$$E[W] = 1.$$

Moreover,

$$P(W = 0) = q,$$

where q is the extinction probability.

The theorem reveals a remarkable connection between the martingale limit and extinction. On the event of survival, the limiting random variable is strictly positive, while convergence to zero occurs precisely when extinction takes place. Thus the martingale limit completely separates the two possible long-term outcomes of the process.
The limiting distribution of W is not arbitrary. By passing to the limit in the generating-function representation of the process, one obtains a functional equation satisfied by its Laplace transform. This equation serves as the starting point for a deeper investigation of asymptotic behaviour and motivates the more refined tools developed in the next section.

The Distribution of the Martingale Limit

The martingale convergence theorem establishes the existence of a limiting random variable W, but it does not determine its distribution. To investigate this question, one studies the Laplace transform

$$\phi(u) = E(e^{-uW}).$$

Recall that

$$E(s^{Z_n}) = f_n(s).$$

Substituting

$$s = e^{-u/m^n},$$

gives

$$E(e^{-uW_n}) = f_n\left(e^{-u/m^n}\right).$$

Using the iteration identity

$$f_n = f \circ f_{n-1},$$

and passing to the limit yields the functional equation

$$\phi(u) = f\left(\phi\left(\frac{u}{m}\right)\right).$$

This equation, often called Abel's functional equation, characterizes the Laplace transform of the martingale limit W. In the supercritical case it provides substantial information about the law of W, but it becomes less useful in the critical and subcritical regimes, where the martingale limit is degenerate.

This observation motivates the search for alternative approaches to asymptotic behaviour. One possibility is to condition on nonextinction, thereby removing the trivial extinction event. The corresponding conditional generating functions are

$$\sum_{k=0}^{\infty} P(Z_n = k \mid Z_n > 0)s^k = \frac{f_n(s) - f_n(0)}{1 - f_n(0)}.$$

The asymptotic behaviour of these generating functions leads naturally to the ratio theorems discussed below.

3.3 Ratio Theorems and Asymptotic Structure

The conditional generating functions introduced above involve ratios of transition probabilities. Consequently, a detailed understanding of asymptotic quantities such as

$$\frac{P_n(i,j)}{P_n(1,1)}$$

becomes essential. The purpose of the ratio theorems is to determine the limiting behaviour of these quantities and thereby provide the technical foundation for the conditioned limit laws developed in the next section.

Throughout this section, let

$$\gamma = f'(q),$$

where q denotes the extinction probability. Whenever the limits exist, define

$$\mathcal{P}(s) = \sum_{j=1}^{\infty} \pi_j s^j.$$

The central idea underlying the ratio theorems is surprisingly simple. By exploiting the positivity properties of derivatives of iterated generating functions, one can establish monotonicity of certain transition-probability ratios. Once monotonicity is known, convergence follows automatically and yields a sequence of coefficients that encode the asymptotic structure of the branching process.

Lemma 3.3 (Monotone Ratio Lemma)**.** *Assume that $p_1 > 0$. Then for every $j \geq 1$,*

$$\frac{P_n(1,j)}{P_n(1,1)} \uparrow \pi_j < \infty.$$

This lemma is the cornerstone of the ratio-theorem theory. The coefficients $\{\pi_j\}$ obtained here serve as the fundamental asymptotic quantities from which all subsequent conditioned limit laws are constructed. Without this result there would be no natural candidates for the limiting coefficients appearing later in the theory.

The limits $\{\pi_j\}$ may be assembled into the generating function

$$\mathcal{P}(s) = \sum_{j=1}^{\infty} \pi_j s^j.$$

Remarkably, these coefficients satisfy a functional equation that links them directly to the offspring generating function.

Theorem 3.4. *The generating function $\mathcal{P}$ satisfies*

$$\mathcal{P}(f(s)) = \gamma \mathcal{P}(s) + \mathcal{P}(p_0),$$

whenever both sides converge.

This equation packages infinitely many relations among the coefficients $\{\pi_j\}$ into a single analytic identity. It may be viewed as an eigenvalue equation for the asymptotic structure of the branching process and becomes the principal object of study in the remainder of the section.

A further analysis shows that $\mathcal{P}(s)$ converges for $0 \leq s < 1$ with convergence at one as well for $m < 1$, and that the above functional equation determines $\mathcal{P}$ uniquely up to a multiplicative constant. Consequently, the coefficients $\{\pi_j\}$ are canonical quantities attached to the branching process and do not depend on arbitrary choices.

The culmination of the theory is the following ratio limit theorem.

Theorem 3.5 (Ratio Theorem)**.** *Under suitable assumptions,*

$$\lim_{n\to\infty} \frac{P_n(k,\ell)}{P_{n+m}(i,j)} = \gamma^m \frac{\pi_k \tau_\ell}{\pi_i \tau_j},$$

where $\{\tau_j\}$ is the corresponding dual sequence.

This theorem provides precise asymptotic comparisons between transition probabilities. It reveals that the long-term behaviour of the process is governed by the coefficients $\{\pi_j\}$ and $\{\tau_j\}$, which act as asymptotic coordinates for the branching mechanism. More importantly, the ratio theorem serves as the principal technical tool for the conditioned limit laws developed in the next section. The limiting distributions obtained there emerge directly from the asymptotic structure uncovered by these ratio limits.

3.4 Conditioning on Survival

In both the subcritical ($m < 1$) and critical ($m = 1$) regimes, extinction occurs with probability one. Consequently,

$$P(Z_n = 0) \longrightarrow 1,$$

and the unconditional distribution of Z_n becomes degenerate. To understand the behaviour of populations that survive for a long time, it is therefore natural to study conditioned distributions.
Let

$$T = \inf\{k \geq 0 : Z_k = 0\}$$

denote the extinction time, and recall the coefficients $\{\pi_j\}_{j\geq 1}$ obtained from the ratio theorem. Define

$$S = \{j \geq 1 : P_n(1, j) > 0 \text{ for some } n\}.$$

When $p_1 > 0$ and $q > 0$, let

$$\widetilde{b}_j = \frac{\pi_j q^j}{\sum_{k=1}^{\infty} \pi_k q^k}, \qquad j \geq 1.$$

If $m \neq 1$, then $\{\widetilde{b}_j\}$ forms a probability distribution on S; when $m = 1$, all coefficients vanish and a different approach is required.
The central question is whether the conditioned distributions

$$P(Z_n = j \,|\, n < T < \infty)$$

possess a nontrivial limit as $n \to \infty$. Here the condition $n < T < \infty$ selects populations that survive up to generation n but are ultimately destined for extinction.

Theorem 3.6. *Assume that $q > 0$. Then*

$$\lim_{n\to\infty} P(Z_n = j \,|\, n < T < \infty) = b_j, \qquad j \geq 1,$$

exists.
Furthermore, if $m \neq 1$, then $\{b_j\}$ is a probability distribution and its generating function

$$B(s) = \sum_{j=1}^{\infty} b_j s^j$$

is the unique solution of

$$B\left(\frac{f(sq)}{q}\right) = \gamma B(s) + (1 - \gamma),$$

among generating functions vanishing at 0.
If $p_1 > 0$, then

$$b_j = \widetilde{b}_j,$$

and

$$B(s) = \frac{\mathcal{P}(qs)}{\mathcal{P}(q)}.$$

Thus a nontrivial limiting distribution emerges after conditioning. The coefficients appearing in the limit are precisely those obtained from the ratio theorem, demonstrating that conditioned limit laws are a direct consequence of the asymptotic structure developed in the previous section.

Corollary 3.7 (Yaglom's Theorem)**.** *If $m < 1$, then*

$$P(Z_n = j \mid Z_n > 0)$$

converges, as $n \to \infty$, to a probability distribution whose generating function satisfies

$$B(f(s)) = mB(s) + (1 - m),$$

Although extinction occurs with probability one in the subcritical regime, the population conditioned on survival converges to a genuine limiting distribution.

The critical case behaves differently. Conditioning only on survival does not produce a proper limiting distribution because the conditioned population continues to grow. A stronger conditioning is therefore required.

Theorem 3.8. *Assume*

$$m = 1, \qquad p_1 > 0.$$

Then

$$\lim_{n\to\infty} P(Z_n = j \mid Z_n > 0,\, Z_{n+1} = 0) = \theta_j,$$

where $\{\theta_j\}_{j\geq 1}$ is a probability distribution with generating function

$$\Theta(s) = \sum_{j=1}^{\infty} \theta_j s^j = \frac{\mathcal{P}(p_0 s)}{\mathcal{P}(p_0)}.$$

This theorem provides a probabilistic interpretation of the coefficients arising from the ratio theorem and highlights the special nature of the critical regime. While conditioned distributions can still be identified, the population size itself continues to grow. Determining the correct normalization for this growth leads to the exponential limit law of the next section.

3.5 The Critical Regime and the Exponential Limit Law

The conditioned limit laws of the previous section show that, in the subcritical case, conditioning on survival is sufficient to produce a nondegenerate limiting distribution. The critical case, however, behaves quite differently. Although conditioning removes the trivial extinction event, the conditioned population continues to grow and does not converge to a proper limiting random variable.

To understand the correct asymptotic scale, it is necessary to examine the survival probability itself. Since extinction occurs with probability one when $m = 1$, the quantity

$$P(Z_n > 0)$$

measures how unlikely it is for a population to survive for n generations. The rate at which this probability decays turns out to determine the typical size of a surviving population.

The conditioned limit laws discussed in the previous section become degenerate in the critical case. Although conditioning on survival removes the trivial extinction event, the population size continues to grow without converging to a proper limiting distribution. Consequently, a deeper asymptotic analysis is required.

The starting point of this analysis is a remarkable estimate describing the rate at which the iterates of the generating function approach their common fixed point. This estimate forms the technical foundation of all subsequent results in the critical case.

Lemma 3.9 (Basic Lemma)**.** *Assume that*

$$m = 1, \qquad \sigma^2 = \mathrm{Var}(Z_1) < \infty.$$

Then

$$\lim_{n\to\infty} \frac{1}{n}\left[\frac{1}{1-f_n(t)} - \frac{1}{1-t}\right] = \frac{\sigma^2}{2},$$

uniformly for $0 \le t < 1$.

The importance of this lemma cannot be overstated. It provides the precise asymptotic rate at which the generating-function iterates approach the fixed point 1. Since the survival probability is given by

$$P(Z_n > 0) = 1 - f_n(0),$$

the lemma immediately yields asymptotic information about survival. In turn, this determines the typical size of a surviving population and ultimately leads to the limiting distribution of the conditioned process. A fundamental estimate due to Kolmogorov establishes the precise asymptotic behavior of the survival probability.

Theorem 3.10 (Kolmogorov Estimate)**.** *Assume that*

$$m = 1, \qquad \sigma^2 = \mathrm{Var}(Z_1) < \infty.$$

Then

$$P(Z_n > 0) \sim \frac{2}{n\sigma^2}.$$

This result reveals that survival becomes increasingly unlikely, but only at a polynomial rate. Unlike the subcritical case, where survival probabilities decay exponentially fast, critical branching processes exhibit a much slower decay. This slow rate is one of the defining features of criticality.
The theorem also identifies the correct scale of the conditioned process. Since

$$E[Z_n] = 1,$$

it follows that

$$E(Z_n \mid Z_n > 0) = \frac{1}{P(Z_n > 0)} \sim \frac{n\sigma^2}{2}.$$

Thus the typical size of a surviving population grows linearly with n. This observation suggests that the appropriate normalization is no longer m^n, as in the supercritical case, but rather n itself.
The remarkable fact is that after this normalization, a universal limiting distribution emerges.

Theorem 3.11 (Yaglom's Exponential Limit Law)**.** *Assume that*

$$m = 1, \qquad \sigma^2 < \infty.$$

Then, for every $z \geq 0$,

$$\lim_{n\to\infty} P\left(\frac{Z_n}{n} > z \,\middle|\, Z_n > 0\right) = \exp\left(-\frac{2z}{\sigma^2}\right).$$

This theorem is one of the most striking results in branching-process theory. After conditioning on survival and scaling by n, the population size converges to an exponential distribution. Even more remarkably, the limiting law depends on the offspring distribution only through its variance. All finer details of the reproduction mechanism disappear in the limit.

Corollary 3.12. *Assume that*

$$m = 1, \qquad \sigma^2 < \infty.$$

Then

$$\lim_{n\to\infty} n^2\big(f_{n+1}(s) - f_n(s)\big) = \frac{2}{\sigma^2}, \qquad s < 1.$$

This result provides a refined description of the rate at which successive iterates of the generating function approach one another. It shows that the asymptotic behavior of the process is sufficiently regular that even the difference between consecutive generations admits a precise limiting form.
Combining the previous corollary with the conditioned limit theorem of the preceding section yields an asymptotic formula for the transition probabilities themselves.

Corollary 3.13. *Assume that*

$$p_1 > 0, \qquad m = 1, \qquad \sigma^2 < \infty.$$

Then, for every $j \geq 1$,

$$\lim_{n\to\infty} n^2 P_n(1, j) = c\, j\, \pi_j,$$

where

$$c = \frac{2}{\sigma^2\, \mathcal{P}(p_0)}.$$

This corollary reconnects the critical theory with the ratio theorems developed earlier. The same coefficients $\{\pi_j\}$, originally introduced through asymptotic ratios of transition probabilities, reappear in the critical limit. This demonstrates the remarkable unity of the theory: the martingale approach, ratio theorems, conditioned limit laws, and exponential limit law ultimately describe different aspects of the same asymptotic structure.

Chapter 4

Strong Convergence in the Supercritical Case

4.1 Strong Convergence in the Supercritical Case

4.1.1 Idea Behind the Convergence

For a supercritical Galton–Watson process, $m = E(Z_1) > 1$, the normalized process is a nonnegative martingale. Hence, by the martingale convergence theorem, $W_n \longrightarrow W \qquad \text{almost surely.}$
The existence of this limit raises the question of whether W contains nontrivial information about the asymptotic growth of the population.
If $E(W) = 1$, then the normalization m^n correctly captures the population growth. However, if $W = 0 \qquad \text{a.s.}$, then the normalization is too large and a slower normalization is required. The sharp criterion separating these two cases is the logarithmic moment $E(Z_1 \log Z_1)$.

4.1.2 Kesten–Stigum Theorem

Assume that $Z_0 = 1, \qquad m > 1,$
and let $W_n = \frac{Z_n}{m^n}$.

Theorem 4.1 (Kesten–Stigum)**.** *The martingale limit W satisfies:*

1. *If $E(Z_1 \log Z_1) < \infty$, then $E(W) = 1$.*

2. *If $E(Z_1 \log Z_1) = \infty$, then $E(W) = 0$.*

Since $W \geq 0$, the second conclusion implies $P(W = 0) = 1$.

Thus the condition $E(Z_1 \log Z_1) < \infty$ is both necessary and sufficient for the classical normalization $\frac{Z_n}{m^n}$ to converge to a nondegenerate limit. If this condition fails, then $\frac{Z_n}{m^n} \longrightarrow 0 \qquad \text{almost surely}$, even though the process may survive forever.

4.1.3 Laplace Transform of the Limit

To study the distribution of W, define $\varphi(u) = E(e^{-uW}), \qquad u \geq 0$. The offspring generating function is $f(s) = \sum_{j=0}^{\infty} p_j s^j$. Using the branching property, the Laplace transform satisfies $\boxed{\varphi(u) = f\left(\varphi\left(\frac{u}{m}\right)\right)}$.
This functional equation characterizes the possible limiting distributions of normalized branching processes.

4.1.4 Analytic Form of the Logarithmic Moment Condition

The proof of the Kesten–Stigum theorem uses an analytic reformulation of the condition $E(X \log X) < \infty$.

Lemma 4.2. *Let $X \geq 0$ with $0 < E(X) < \infty$. Then, for $a > 0$, $\int_0^a \frac{E(e^{-uX/m} - e^{-u})}{u^2}\, du < \infty$ if and only if $E(X \log X) < \infty$.*

For a Galton–Watson offspring distribution this becomes the equivalent condition $\int_0^1 \frac{f(e^{-u/m}) - e^{-u}}{u^2}\, du < \infty$ if and only if $\sum_{j=0}^{\infty} p_j j \log j < \infty$. Hence the logarithmic moment condition can be studied through the generating function itself.

4.1.5 Seneta Normalization

The Kesten–Stigum theorem shows that when $E(Z_1 \log Z_1) = \infty$, the classical normalization fails: $\frac{Z_n}{m^n} \longrightarrow 0 \quad$ a.s. However, this does not imply that no nontrivial limit exists. Seneta showed that there exists a slower normalization sequence $\{C_n\}$ such that $C_n^{-1} Z_n \longrightarrow W^*$ a.s., where W^* is nondegenerate.

Theorem 4.3 (Seneta's Normalization)**.** *Let $1 < m < \infty$. There exists a sequence $\{C_n\}$ satisfying $C_n \to \infty$, and $\frac{C_n}{C_{n+1}} \longrightarrow m^{-1}$, such that $W_n^* = \frac{Z_n}{C_n}$ converges almost surely to a random variable W^* with $P(W^* > 0) > 0$. Moreover, the Laplace transform $\phi(u) = E(e^{-uW^*})$ satisfies*

$$\boxed{\phi(u) = f\left(\phi\left(\frac{u}{m}\right)\right)}.$$

Thus the failure of m^n is due only to an incorrect scaling; a modified normalization still produces a meaningful limiting distribution.

4.1.6 Construction of the Normalization

Define $k(s) = -\log f(e^{-s})$, which is increasing, continuous, and concave. Let h denote its inverse function. For a suitable s_0, $C_n = \frac{1}{h_n(s_0)}$, where h_n denotes the n-fold iterate of h. The construction leads to the transformed process $Y_n = e^{-W_n^*}$, which forms a bounded martingale.

Lemma 4.4. *The sequence $\{Y_n\}$ satisfies $Y_n \longrightarrow Y \qquad$ a.s. and $W^* = -\log Y$.*

Furthermore, $P(Y > 0) = 1$, and $P(Y = 1) = q$, where q is the extinction probability. Hence the limit variable retains the extinction-survival structure of the original process.

4.1.7 Uniqueness of the Functional Equation

The previous results show that limiting distributions satisfy $\phi(u) = f\left(\phi\left(\frac{u}{m}\right)\right)$. The next theorem studies uniqueness of solutions to this equation.
Let $\mathcal{C}$ denote the class of Laplace transforms $\phi(u) = \int_0^\infty e^{-ux}\, dF(x)$, where F is a nondegenerate distribution on $[0, \infty)$.

Theorem 4.5. *The functional equation $\phi(u) = f\left(\phi\left(\frac{u}{m}\right)\right)$ has a unique solution in $\mathcal{C}$, up to a change of scale.*
Moreover, $\int_0^\infty t^p |\log t|^p\, dF(t) < \infty$ if and only if $\sum_{j=0}^{\infty} p_j j (\log j)^{p+1} < \infty$.

This establishes that the offspring distribution completely determines the possible moment properties of the limiting distribution.

Corollary 4.6. *If* $F(0+) = q$, *then* $P(W = 0) = q$.

Therefore, the limit variable separates the two possibilities: $W = 0 \quad$ (extinction), and $W > 0 \quad$ (survival).

4.2 Geometric Convergence of $f_n(s)$ in the Noncritical Cases

4.2.1 Motivation

The iterated offspring generating function $f_n(s) = \underbrace{f(f(\cdots f(s)))}_{n \text{ times}}$, where $f(s) = \sum_{j=0}^{\infty} p_j s^j$, satisfies $f_n(s) = E(s^{Z_n})$. Hence, the asymptotic behaviour of $f_n(s)$ describes the limiting behaviour of the population. Since $f_n(s) \longrightarrow q$, where q is the extinction probability satisfying $f(q) = q$, the main question is the rate at which this convergence occurs. The behaviour differs fundamentally from the critical case. In the noncritical regimes, the convergence is geometric and is controlled by the derivative of f at the fixed point.

4.2.2 Scaled Convergence of the Iterates

Let $\gamma = f'(q)$. The quantity $Q_n(s) = \gamma^{-n}(f_n(s) - q)$ rescales the difference between the iterates and their limiting fixed point.

Theorem 4.7. *For* $0 \le s < 1$, $Q_n(s) \longrightarrow Q(s)$ *as* $n \to \infty$.
Moreover, if $m < 1$ *and* $\sum_j p_j j \log j = \infty$, *then* $Q'(s) = 0, \qquad 0 \le s \le 1$.
In all other cases, $Q'(s) > 0, \qquad 0 \le s < 1$, *and* $\lim_{s \uparrow q} Q'(s) = 1$.

Thus the convergence has the asymptotic form $f_n(s) - q \sim \gamma^n Q(s)$. The appearance of $\gamma = f'(q)$ follows from the local approximation $f(s) - q \approx f'(q)(s - q)$ near the fixed point q. Hence repeated iteration produces the geometric factor γ^n.
The proof is based on differentiating the iterates: $Q_n'(s) = \prod_{j=0}^{n-1} \frac{f'(f_j(s))}{\gamma}$, and studying the convergence of the resulting infinite product. The condition involving $\sum_j p_j j \log j$ controls the approach of $f'(f_j(s))$ towards $f'(q)$.
The limiting function is normalized by $Q(s) = \int_q^s Q'(x)\, dx$, so that $Q(q) = 0$.

Corollary 4.8. *If* $m \neq 1$, *then* $\lim_{n\to\infty} \gamma^{-n}(f_n(s) - q) = Q(s), \qquad 0 \le s < 1$.

The function Q gives the first-order asymptotic behaviour of the iterated generating functions.

4.2.3 Functional Equation for the Limit Function

The limiting function satisfies a linearizing relation.

Theorem 4.9. *The function* Q *satisfies* $\boxed{Q(f(s)) = \gamma Q(s)}$ *for* $0 \le s < 1$, *together with* $Q(q) = 0$, *and* $\lim_{s\uparrow q} Q'(s) = 1$.

This equation converts the nonlinear iteration of f into a linear multiplication by γ. It is analogous to Schröder's equation in dynamical systems and shows that Q acts as a linearizing coordinate for the branching process.

4.2.4 Connection with Transition Probabilities

The generating function asymptotics also determine the asymptotic transition probabilities.

Theorem 4.10. *For fixed $i, j \geq 1$, $\lim_{n\to\infty} \gamma^{-n} P_{ij}^{(n)} = iq^{i-1} v_j$, where v_j are the coefficients in the expansion $Q(s) = \sum_{j=0}^{\infty} v_j s^j$.*

The proof follows by using $\sum_j s^j P_{ij}^{(n)} = f_n(s)^i$ and substituting the expansion $f_n(s) = q + \gamma^n Q(s) + o(\gamma^n)$. Thus the coefficients of the limiting analytic function Q determine the asymptotic profile of the transition probabilities.

4.2.5 Application: Conditional Behaviour in the Subcritical Case

Although a subcritical process satisfies $Z_n \longrightarrow 0$, conditioning on the rare event $Z_n > 0$ reveals a non-trivial limiting population structure.

Theorem 4.11 (Joffe–Spitzer)**.** *Assume $m < 1$ and $\sum_j jp_j \log j < \infty$. Then $P(Z_n = k \mid Z_n > 0) \longrightarrow d_k(c)$, where $d_k(c) \geq 0$, and $\sum_k d_k(c) = 1$. The generating function of the limiting distribution is $\sum_k d_k(c) s^k = e^{-c(1-B(s))}$.*

This result shows that although extinction occurs with probability one, populations conditioned on long survival possess a stable limiting structure. It forms one of the foundations of conditioned limit theory and leads naturally to Yaglom-type results.

Chapter 5

The Q-Process and Diffusion Limits

5.1 The Q-Process

5.1.1 Motivation

In the critical and subcritical cases, $m \leq 1$, a Galton–Watson process becomes extinct almost surely. If $T = \inf\{n : Z_n = 0\}$, then $P(T < \infty) = 1$. However, many questions concern populations that survive for an unusually long time. Conditioning on $\{n < T < \infty\}$ only describes survival up to a finite time. A more natural conditioning is $\{T > n + k\}$, which asks that extinction is delayed far into the future. The limiting process obtained from this conditioning is called the Q-process.

5.1.2 Construction through Long Survival Conditioning

Consider the conditional distribution

$$P(Z_n = j \mid n + k < T < \infty).$$

Using the Markov property, this probability can be expressed in terms of the transition probabilities

$$P_n(i, j) = P(Z_n = j \mid Z_0 = i).$$

Taking the limit $k \to \infty$ produces a new limiting transition structure.

Theorem 5.1 (Existence of the Q-Process). *Assume $p_1 > 0$ and let q denote the extinction probability. Then*

$$\lim_{k\to\infty} P(Z_n = j \mid n + k < T < \infty) = b_j(k)$$

exists.
For $m \neq 1$,

$$b_j(k) \geq 0, \qquad \sum_{j\geq 1} b_j(k) = 1.$$

In the critical case $m = 1$, the corresponding limiting distribution is denoted by $\theta_j(k)$ and satisfies

$$\sum_{j\geq 1} \theta_j(k) = 1.$$

Thus, conditioning on very long survival does not merely modify probabilities; it produces a new stochastic process biased towards survival.

5.1.3 The Q-Process as a Doob h-Transform

The limiting conditional probabilities define a Markov chain $\{\widetilde{Z}_n : n \geq 0\}$, called the Q-process. Its transition probabilities are given by

$$Q_n(i,j) = P_n(i,j)\frac{jq^{j-i}}{i}, \qquad i,j \geq 1,$$

with the normalization determined by the limiting harmonic function.
This transformation is an example of a Doob h-transform.
Although the Q-process is often interpreted as a branching process conditioned on never becoming extinct, the event $\{T = \infty\}$ has probability zero. Hence the Q-process is obtained through limiting conditioning rather than direct conditioning.

5.1.4 Classification of the Q-Process

After constructing the process, the next question concerns its long-term Markov behaviour.

Theorem 5.2. *The recurrence properties of the Q-process are determined by the offspring mean.*

1. *If $m > 1$, then the Q-process is positive recurrent.*
2. *If $m = 1$, then the Q-process is transient.*
3. *If $m < 1$, then the Q-process is positive recurrent if and only if $\sum_j p_j j \log j < \infty$.*

In the positive recurrent cases, the stationary distribution is

$$\mu_j = jq^{j-1}v_j, \qquad j \geq 1,$$

where

$$Q(s) = \sum_{j\geq 0} v_j s^j$$

is the limiting function obtained from the geometric convergence of generating functions.

The proof follows from the asymptotic transition probabilities

$$P_n(i,j) \sim \gamma^n i q^{i-1} v_j,$$

which yield the limiting invariant measure of the transformed process.

5.1.5 Critical Q-Process and Scaling

The critical case is particularly interesting. Although $m = 1$ implies $Z_n \longrightarrow 0 \quad$ a.s., conditioning on infinite survival produces a growing population. The correct scale is linear in n.

Theorem 5.3. *Assume $m = 1$ and $\sigma^2 = \mathrm{Var}(Z_1) < \infty$. Then $\frac{\sigma^2}{2n}\widetilde{Z}_n$ converges in distribution to a random variable with density*

$$h(x) = xe^{-x}, \qquad x > 0.$$

The limiting density is the Gamma distribution $\Gamma(2, 1)$. Hence the critical Q-process grows as $\widetilde{Z}_n = O(n)$, in contrast with the ordinary critical process, which becomes extinct almost surely.

5.2 More on Conditioning and Limiting Diffusions

5.2.1 Motivation

The Q-process describes a Galton–Watson process conditioned on surviving indefinitely. A related question is to understand the behaviour of a critical branching process conditioned only on survival up to a large finite time. This leads naturally to diffusion approximations, where the discrete branching process converges to a continuous-state stochastic process.

5.2.2 Conditioning on Survival to a Fixed Time

Throughout this section, let $m = 1$, and let $T = \inf\{n : Z_n = 0\}$. Although $E(Z_n) = 1$, the process satisfies $P(T < \infty) = 1$. Thus survival up to generation n is a rare event. The appropriate object to study is the conditioned process $\{Z_{[nt]} \mid Z_n > 0\}, \qquad 0 \le t \le 1$. Since surviving populations are typically of order n, the correct scaling is $\frac{Z_{[nt]}}{n}$.

Theorem 5.4 (Spitzer; Lamperti–Ney)**.** *Assume*

$$m = 1, \qquad \sigma^2 = f''(1) < \infty.$$

Then, for every $0 < t < 1$,

$$\left.\frac{Z_{[nt]}}{n}\ \right|\ Z_n > 0$$

converges in distribution to

$$U + V,$$

where U and V are independent exponential random variables with parameters

$$U \sim \mathrm{Exp}\left(\frac{2}{\sigma^2 t}\right), \qquad V \sim \mathrm{Exp}\left(\frac{2}{\sigma^2(1-t)}\right).$$

Thus, conditioned on survival, the population grows linearly with n, in contrast to the unconditioned process, which converges to extinction almost surely. The limiting population may be viewed as the combined contribution of past growth and the future requirement of survival.

5.2.3 Relation with the Q-Process

The conditioned process

$$\{Z_n \mid Z_n > 0\}$$

and the Q-process arise from different conditioning procedures. The former conditions on survival up to a fixed generation, while the latter conditions on survival indefinitely. Both constructions transform rare survival events into nontrivial limiting stochastic processes and are closely related through their asymptotic behaviour.

5.2.4 Diffusion Approximation

To study the evolution of the entire population trajectory, consider the rescaled process

$$X_n(t) = \frac{Z_{[nt]}}{n}, \qquad 0 \le t \le 1.$$

Under this scaling, the discrete branching process converges to a continuous diffusion process. The limiting diffusion satisfies

$$dX_t = \sigma\sqrt{X_t}\,dB_t,$$

where B_t is standard Brownian motion and σ^2 is the offspring variance.
This process is known as the Feller branching diffusion. The diffusion coefficient is proportional to $\sqrt{X_t}$, reflecting the fact that larger populations experience greater fluctuations while extinction remains an absorbing state.
The Q-process admits an analogous scaling. The rescaled process

$$\frac{\widetilde{Z}_{[nt]}}{n}$$

converges to the corresponding branching diffusion conditioned to remain positive, providing the continuous analogue of conditioning on eternal survival.

5.2.5 Summary

The results of this chapter establish three complementary viewpoints for studying branching processes.

- The ordinary process describes the actual population dynamics and ultimately becomes extinct in the critical case.
- Conditioning on survival up to a large finite time produces nontrivial Yaglom-type limits and linear population growth.
- Conditioning on indefinite survival yields the Q-process, whose large-scale behaviour converges to a conditioned Feller branching diffusion.

Together, these results illustrate how conditioning and scaling transform a discrete branching process into continuous stochastic models while preserving its essential probabilistic structure.